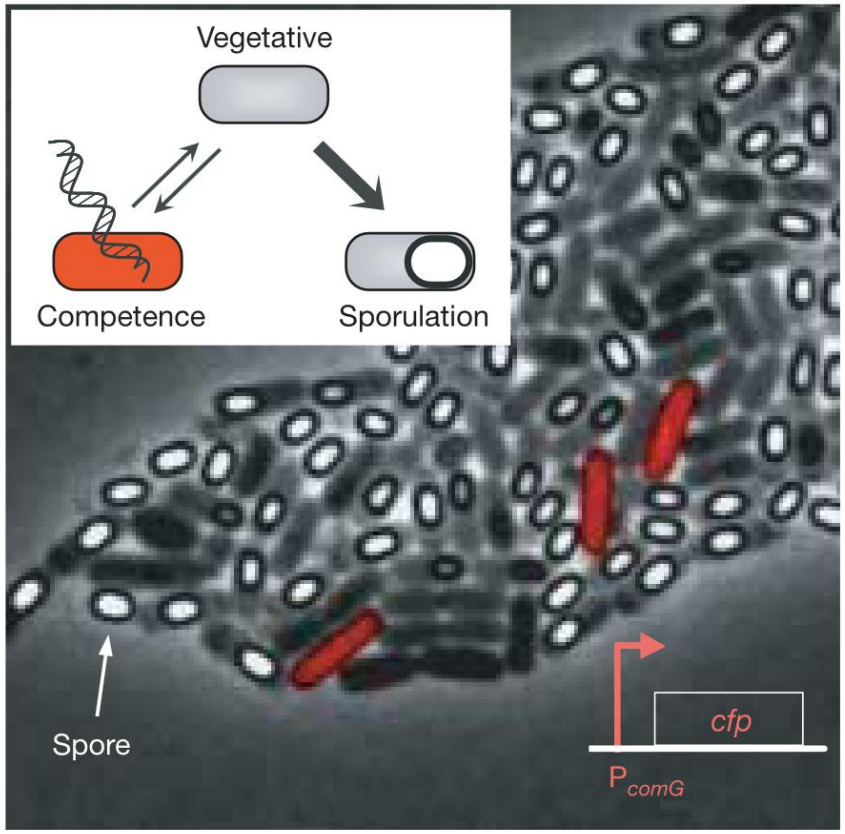
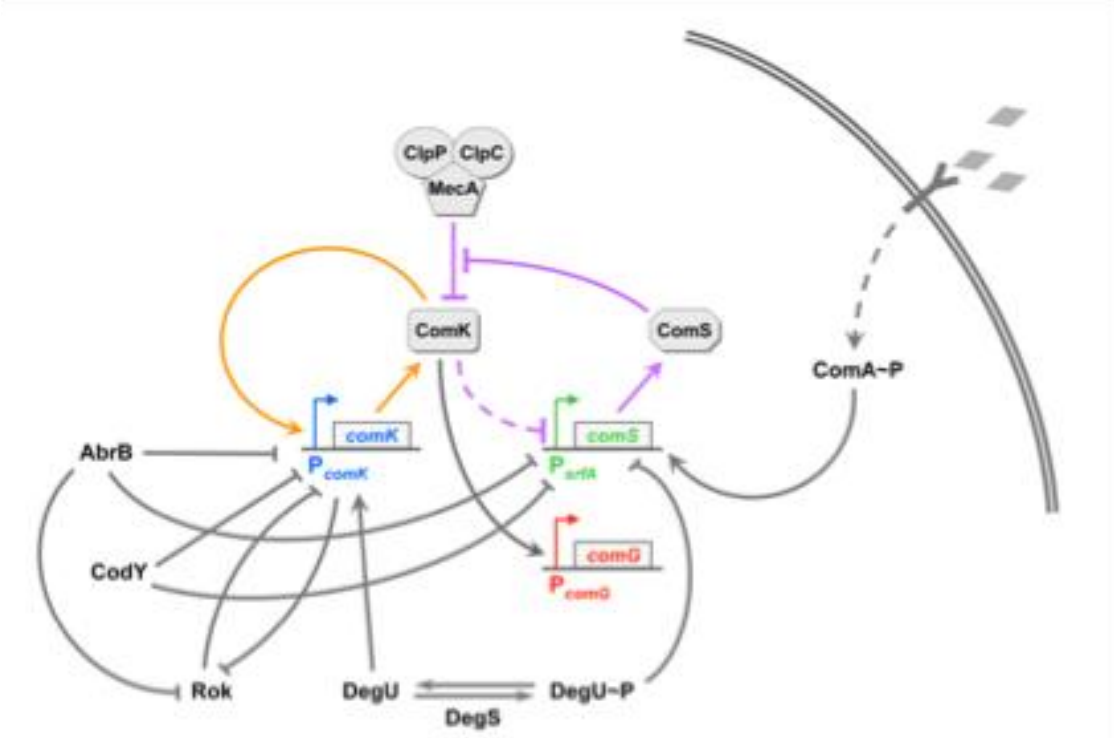
Competence is transient developmental fate that a minority of cells enter



P(Comp) = 3.6 ± 0.7%

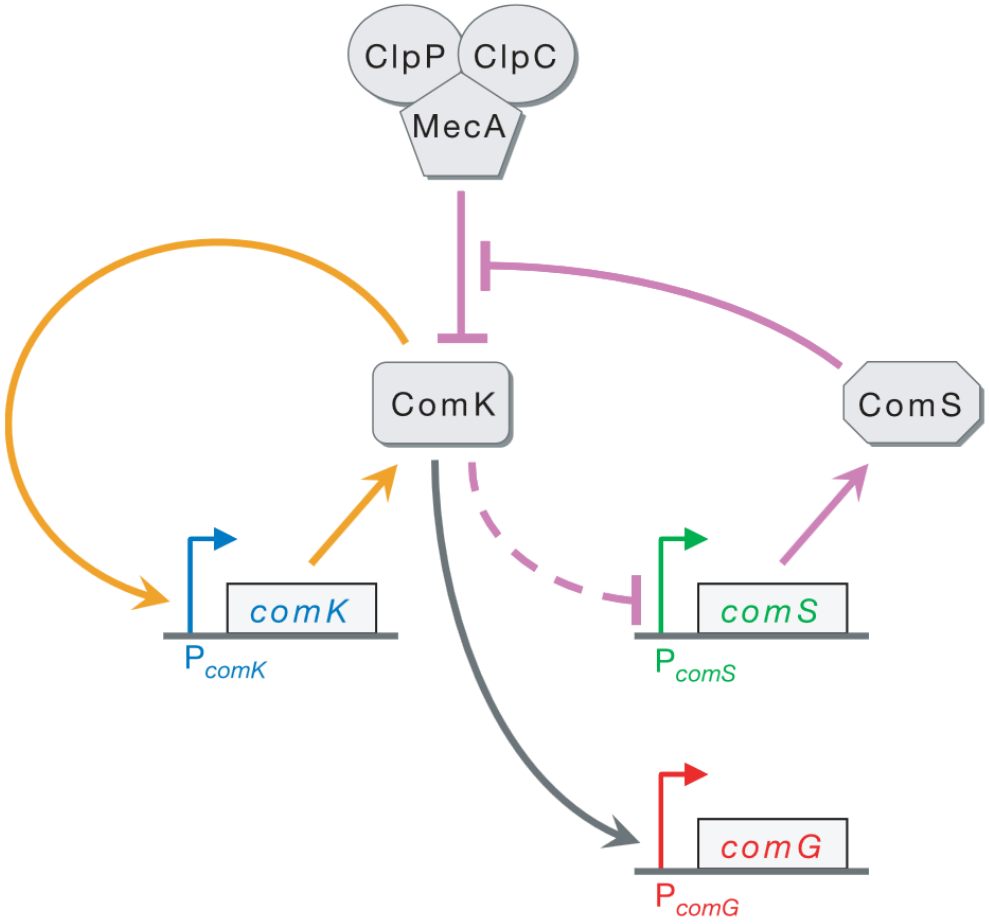
Süel GM et al, Nature (2006)

Extensive prior genetics and biochemical work have mapped out the genetic circuit of competence amidst a dense GRN



Süel GM et al, Nature (2006)

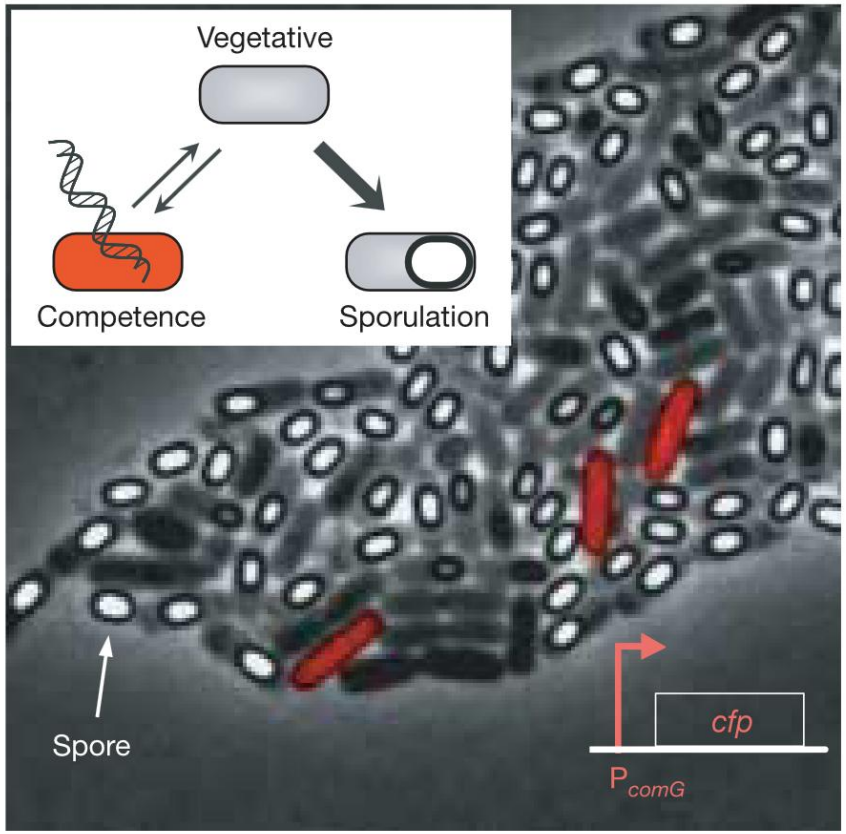
MeKS module entails comKS and MecA which are previously implicated in competence

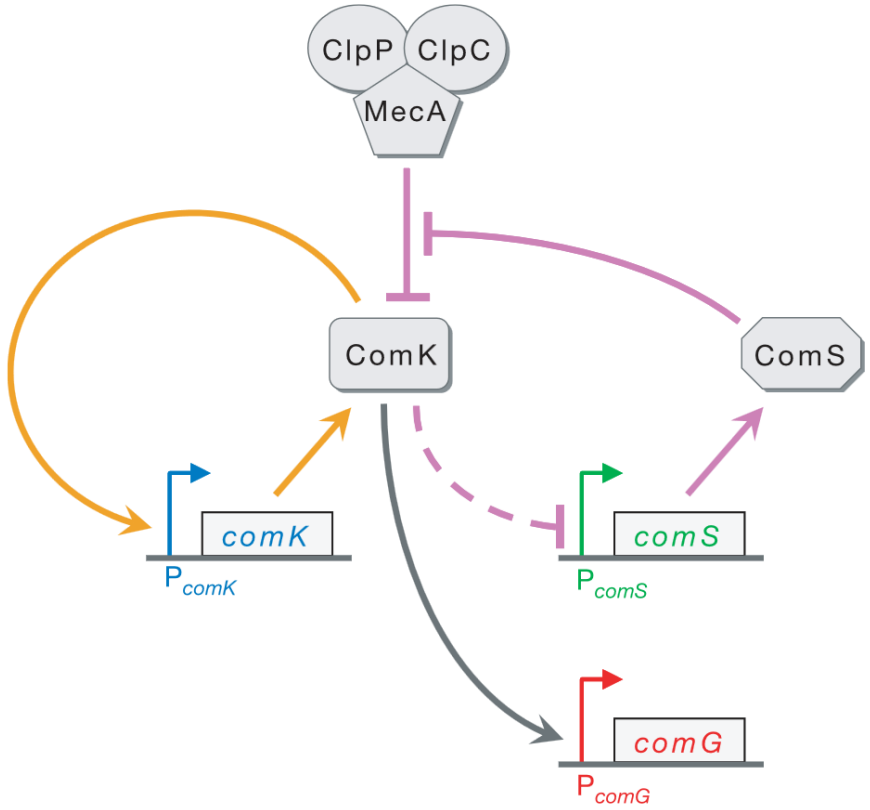


Süel GM et al, Nature (2006)

Traditional bulk based experimental methods are unsuited for studying minority, transient and non-synchronous phenotypes

The authors constructed pair-wise combinations of P_{comK}, P_{comG} and P_{comS} expressing yfp and cfp

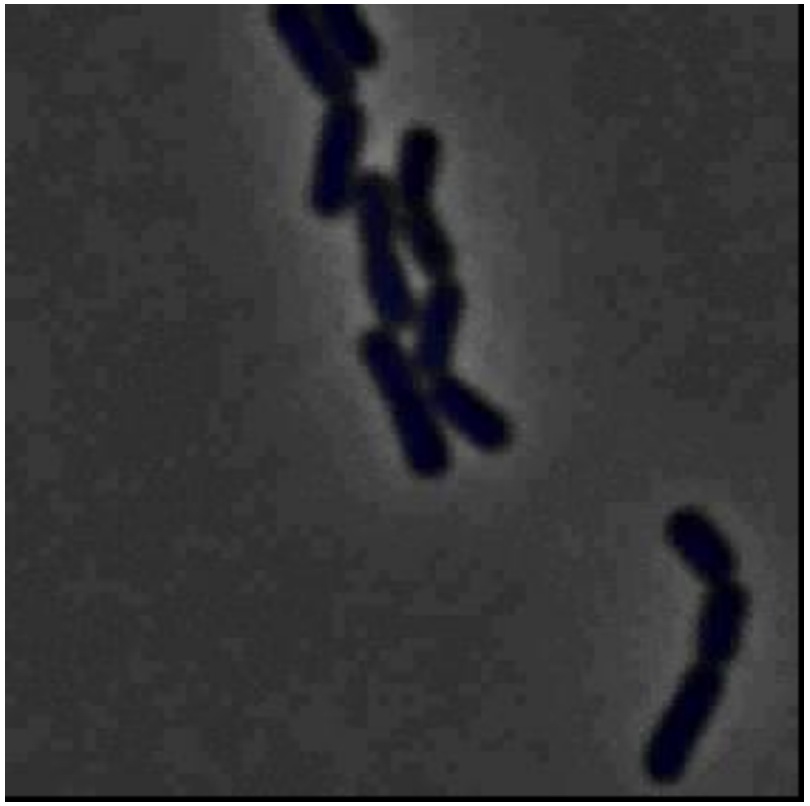


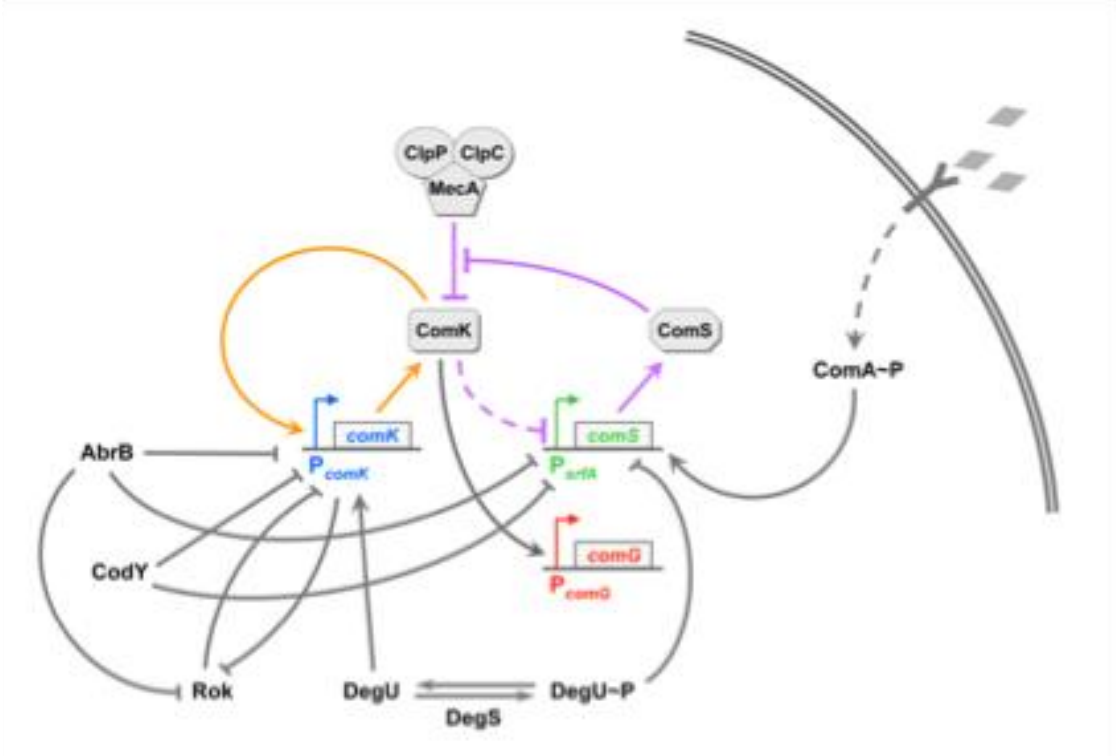


Süel GM et al, Nature (2006)

The first question they ask is whether the system can be reduced to the core circuit

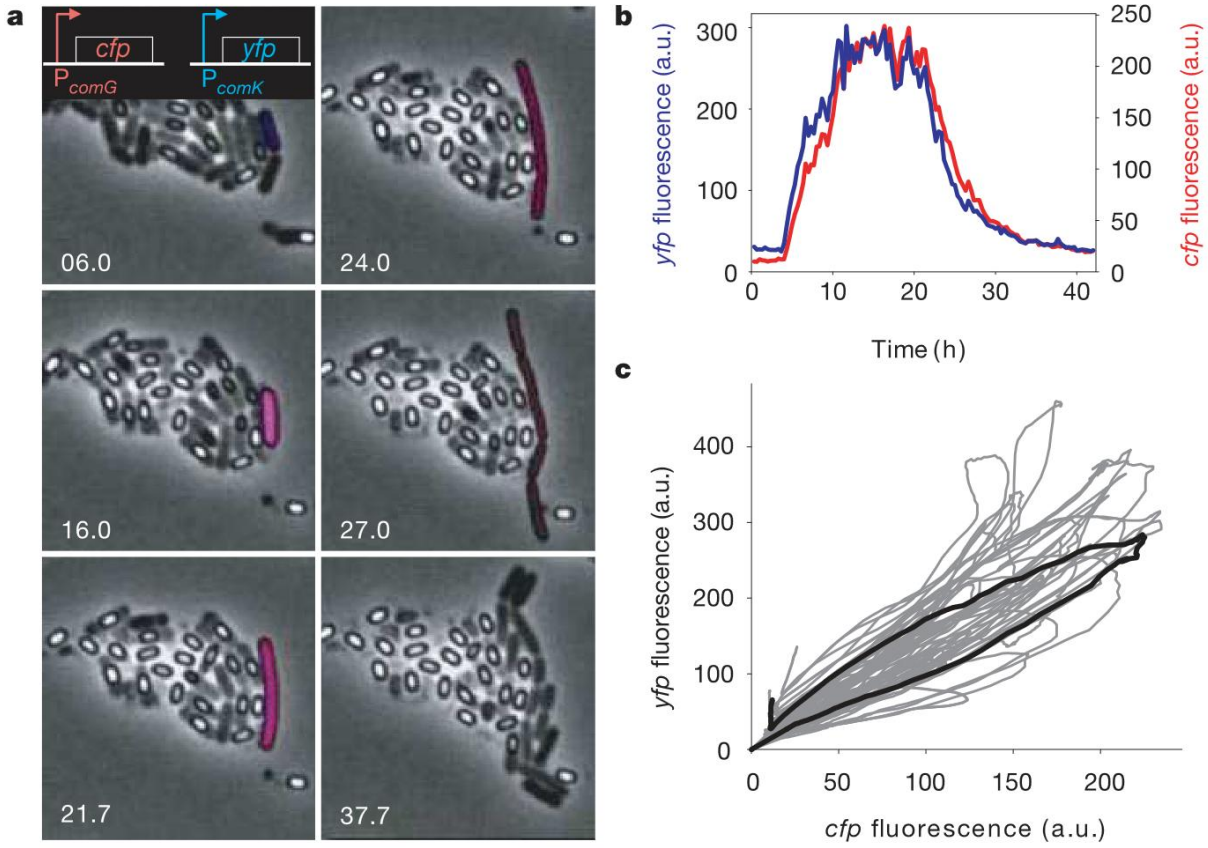


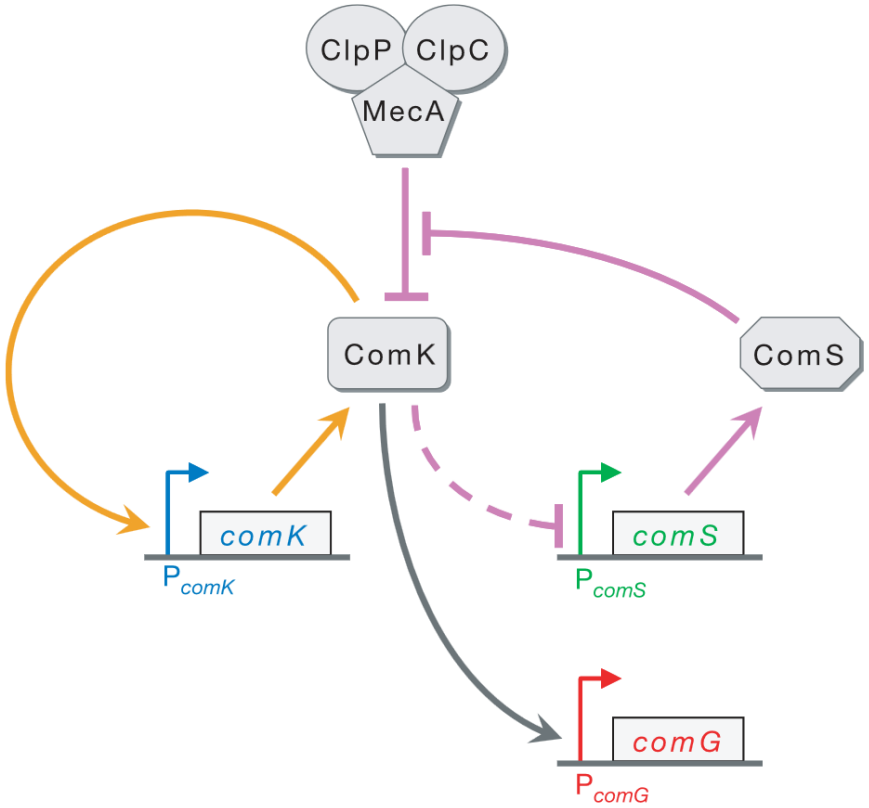




Süel GM et al, Nature (2006)

High degree of correlation between P_{comG} and P_{comK} promoters suggests minimal other inputs for ComK during competence

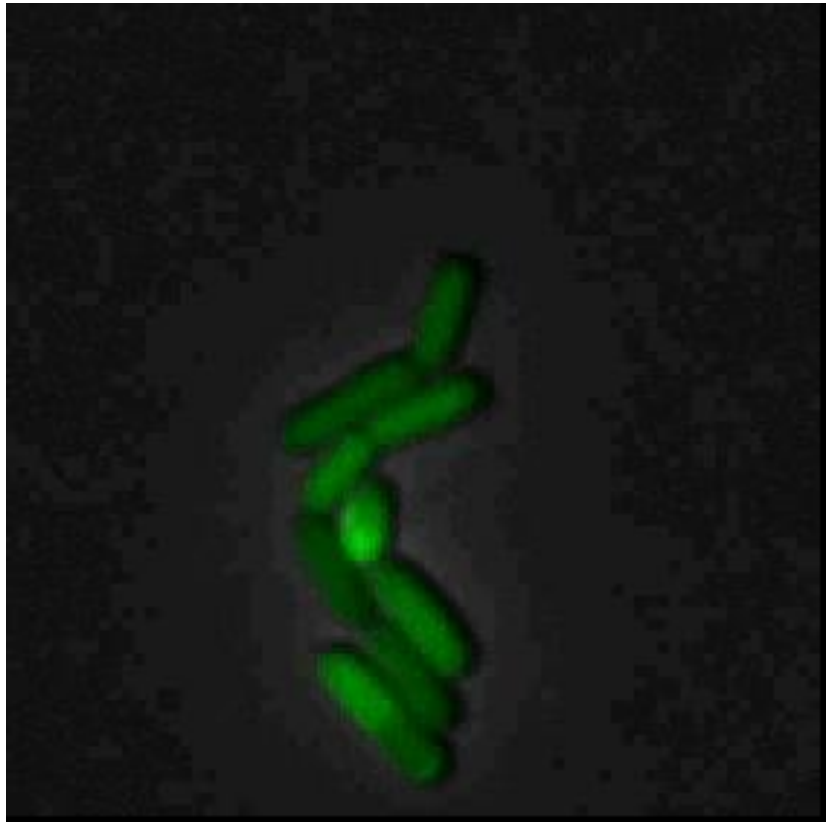


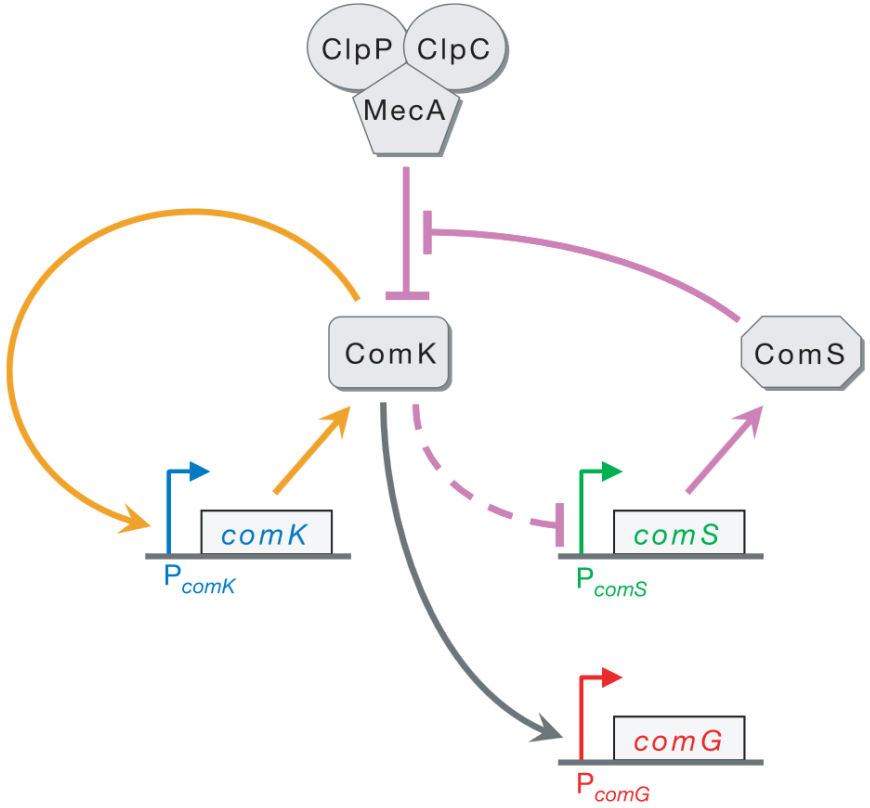


Süel GM et al, Nature (2006)

P_{comG} and P_{comS} transcription activities are anti-correlated during competence

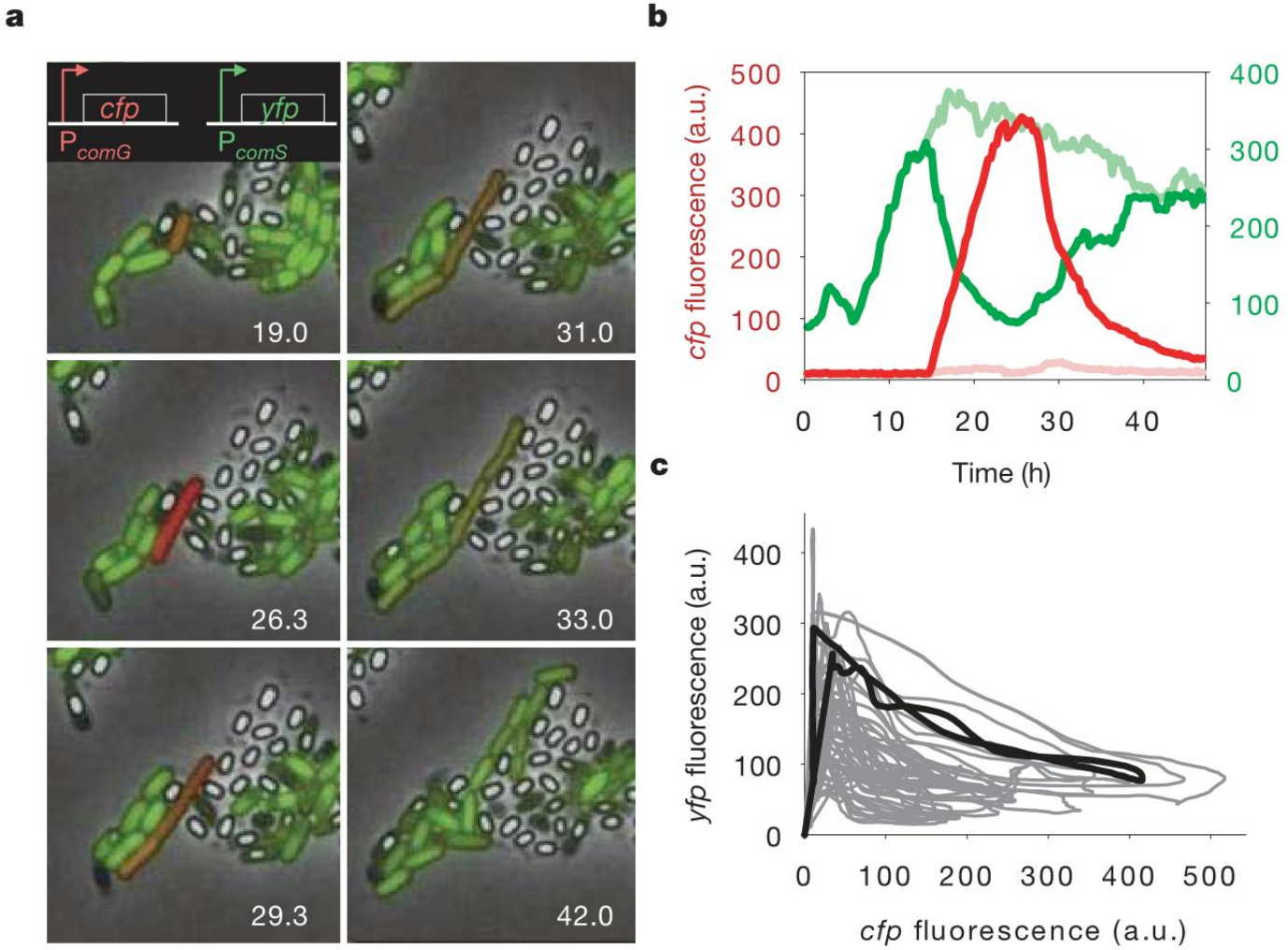






Süel GM et al, Nature (2006)

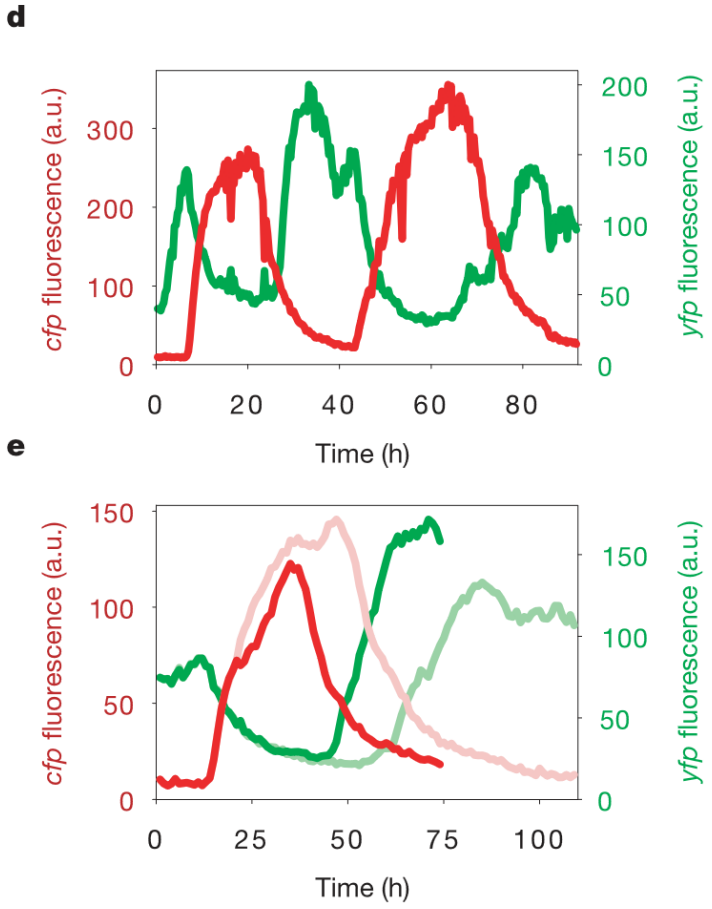
Anti-correlation between P_{comG} and P_{comS} supports ComS-Mediated Negative Feedback



Süel GM et al, Nature (2006)

Repeated competence events and sister-cell variability support stochastic Initiation

P(Comp) = 3.6 ± 0.7%

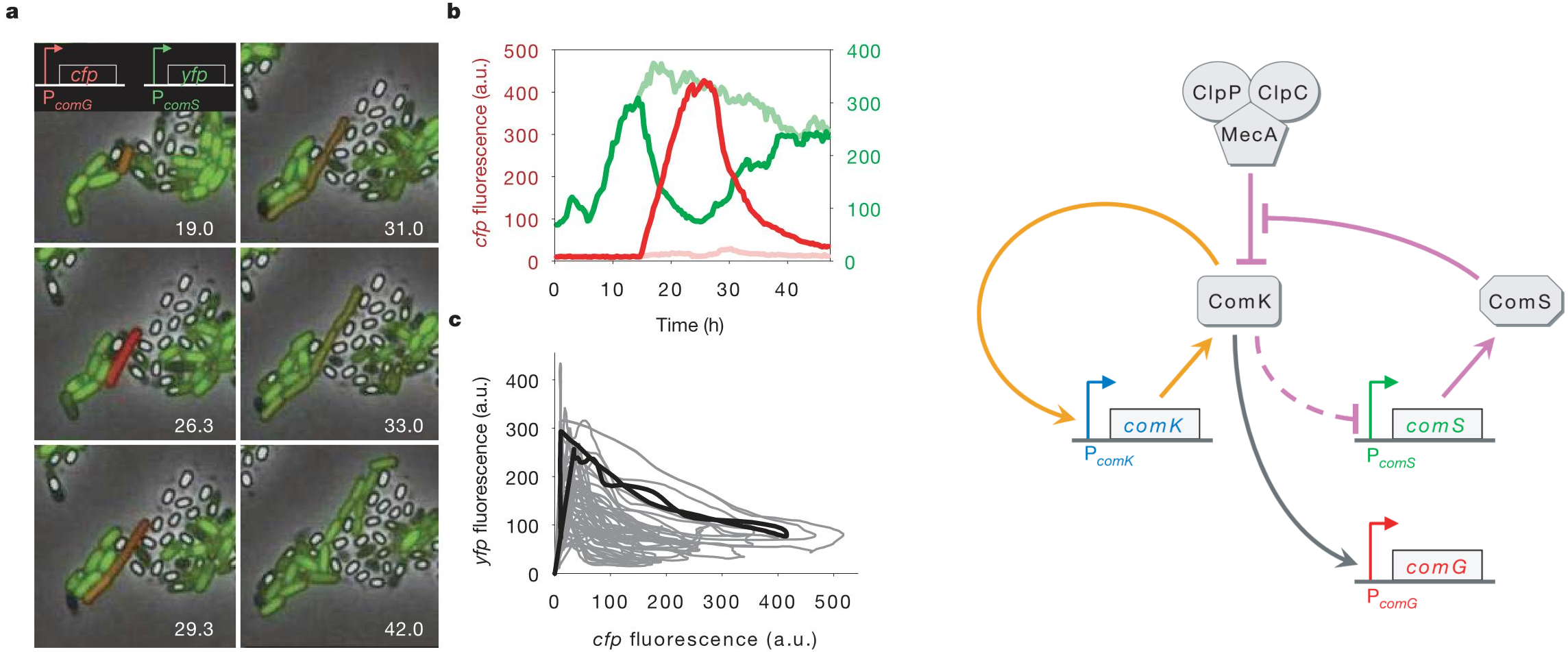


P(Comp | Comp in the past) = 6.0 ± 2.0%

P(Comp | Sister Comp) = 4.1 ± 0.9%

Süel GM et al, Nature (2006)

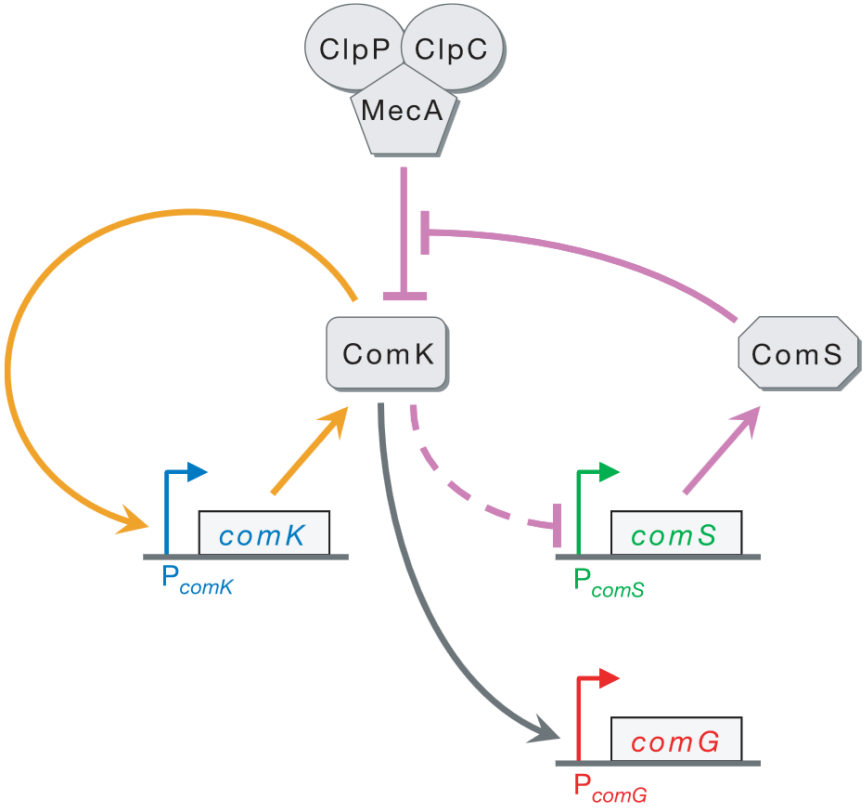
How do the positive+negative feedback circuit cause stochasticity?



Süel GM et al, Nature (2006)

Why doesn’t the system oscillate or stay bistable?

The authors proposed a 2D ODE system of ComKS to explain the experimental observed behavior



$$\frac{dK}{dt} = \alpha_k + \frac{\beta_k K^n}{k_k^n + K^n} - \frac{\delta_k K}{1 + K/\Gamma_k + S/\Gamma_s}$$

$$\frac{dS}{dt} = \frac{\beta_s}{1 + (K/k_s)^p} - \frac{\delta_s S}{1 + K/\Gamma_k + S/\Gamma_s}$$

Süel GM et al, Nature (2006)

Biochemically interpretable parameters allow the model to be anchored by measurements and plausible estimates

$$\frac{dK}{dt} = \alpha_k + \frac{\beta_k K^n}{k_k^n + K^n} - \frac{\delta_k K}{1 + K/\Gamma_k + S/\Gamma_s}$$

$$\frac{dS}{dt} = \frac{\beta_s}{1 + (K/k_s)^p} - \frac{\delta_s S}{1 + K/\Gamma_k + S/\Gamma_s}$$

Parameter	Description
α_k	Basal expression rate of ComK
β_k	Saturating expression rate of ComK positive feedback
β_s	Unrepressed maximal expression rate of ComS
k_k	ComK concentration for half-maximal ComK activation
k_s	ComK concentration for half-maximal ComS repression
δ_k	Unrepressed degradation rate of ComK
δ_s	Unrepressed degradation rate of ComS
Γ_k	ComK concentration for half-maximal degradation
Γ_s	ComS concentration for half-maximal degradation
n	Hill coefficient of ComK positive feedback
p	Hill coefficient of ComS repression by ComK

Süel GM et al, Nature (2006)

Nondimensionalization reduces the model while preserving interpretable control parameters

$$\frac{dK}{dt} = \alpha_k + \frac{\beta_k K^n}{k_k^n + K^n} - \frac{\delta_k K}{1 + K/\Gamma_k + S/\Gamma_s}$$
$$\frac{dS}{dt} = \frac{\beta_s}{1 + (K/k_s)^p} - \frac{\delta_s S}{1 + K/\Gamma_k + S/\Gamma_s}$$

$\delta_k^{-1} = \delta_s^{-1} = 0.0014 s^{-1}$

$\Gamma_k = 500nM$ and $\Gamma_s = 50nM$

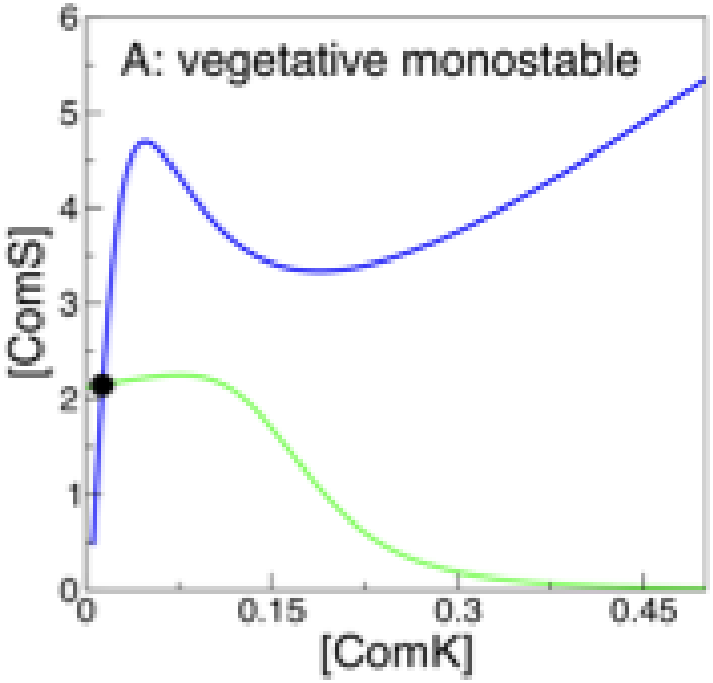
$$\frac{dK}{dt} = a_k + \frac{b_k K^n}{k_0^n + K^n} - \frac{K}{1 + K + S}$$
$$\frac{dS}{dt} = \frac{b_s}{1 + (K/k_1)^p} - \frac{S}{1 + K + S}$$

Parameter	Description
α_k	Basal expression rate of ComK
β_k	Saturating expression rate of ComK positive feedback
β_s	Unrepressed maximal expression rate of ComS
k_k	ComK concentration for half-maximal ComK activation
k_s	ComK concentration for half-maximal ComS repression
δ_k	Unrepressed degradation rate of ComK
δ_s	Unrepressed degradation rate of ComS
Γ_k	ComK concentration for half-maximal degradation
Γ_s	ComS concentration for half-maximal degradation
n	Hill coefficient of ComK positive feedback
p	Hill coefficient of ComS repression by ComK

Parameter	**Definition**	**Value**
a_k	$\alpha_k/(\delta_k\Gamma_k)$	0.004
b_k	$\beta_k/(\delta_k\Gamma_k)$	0.07
b_s	$\beta_s/(\delta_s\Gamma_s)$	0.82
k_0	k_k/Γ_k	0.2
k_1	k_s/Γ_k	0.222
n	–	2
p	–	5

Süel GM et al, Nature (2006)

The dynamics can be read from the ComK and ComS nullclines



$$\dot{K} = 0: \qquad S = \frac{K}{a_k + \dfrac{b_k K^n}{k_0^n + K^n}} - 1 - K$$

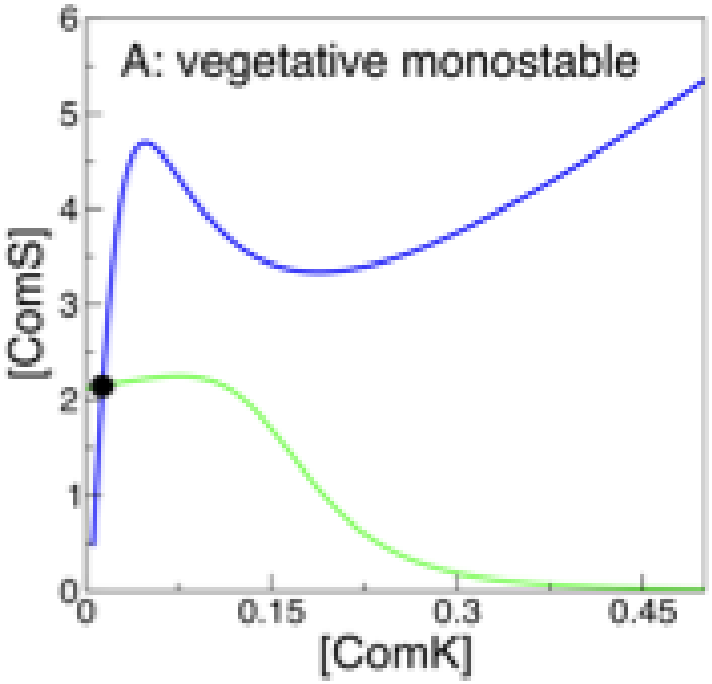
$$\dot{S} = 0: \qquad S = \frac{q(K)(1+K)}{1-q(K)}, \qquad q(K) = \frac{b_s}{1+(K/k_1)^p}$$

Parameter	Meaning	Value
a_k	Basal expression rate of ComK	0.004
b_k	Saturating expression rate of ComK positive feedback	0.07
b_s	Unrepressed maximal expression rate of ComS	0.82
k_0	ComK concentration for half-maximal ComK activation	0.20
k_1	ComK concentration for half-maximal ComS repression	0.222
n	Hill coefficient of ComK positive feedback	2
p	Hill coefficient of ComS repression by ComK	5

Süel GM et al, Nature (2006)

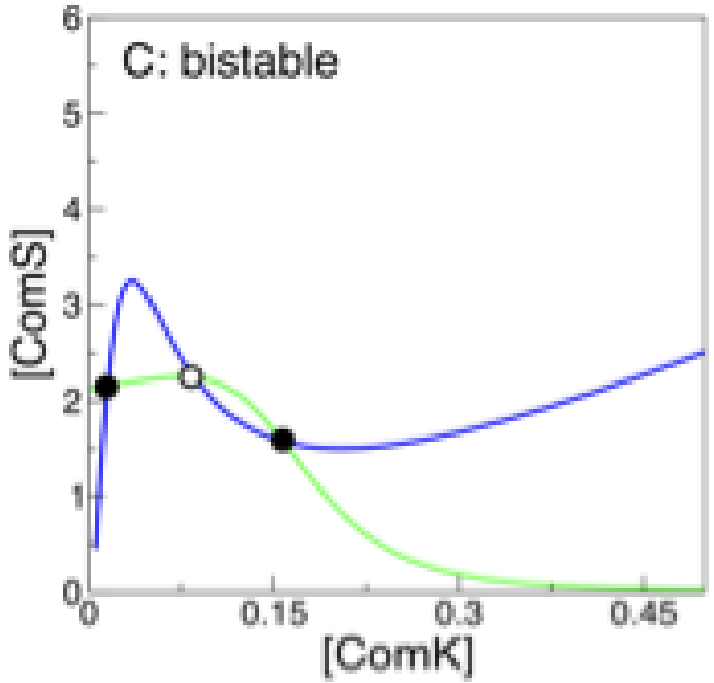
Increasing positive feedback strength (b_k) convert monostability to bistability through a saddle-node bifurcation

$$\dot{K} = 0: \qquad S = \frac{K}{a_k + \frac{b_k K^n}{k_0^n + K^n}} - 1 - K$$



$b_k : 0.08 \rightarrow 0.14$

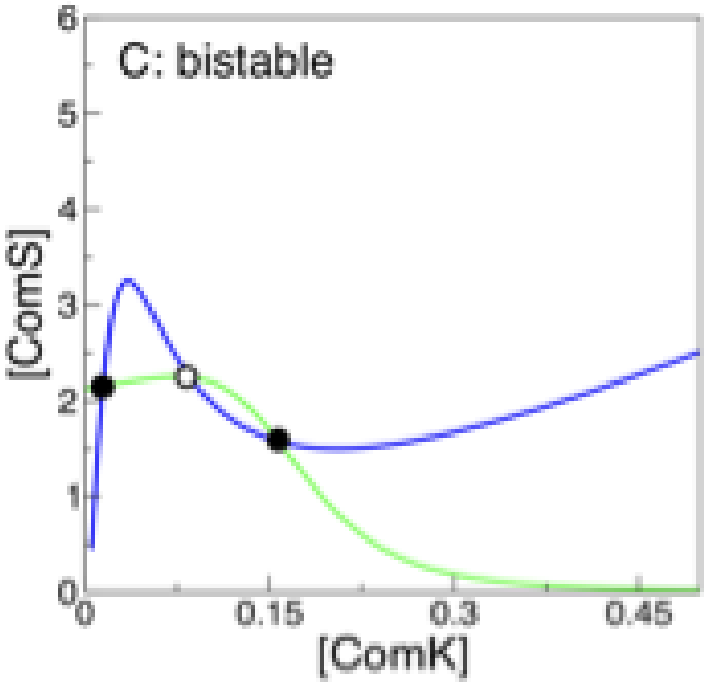
$b_s = 0.68$



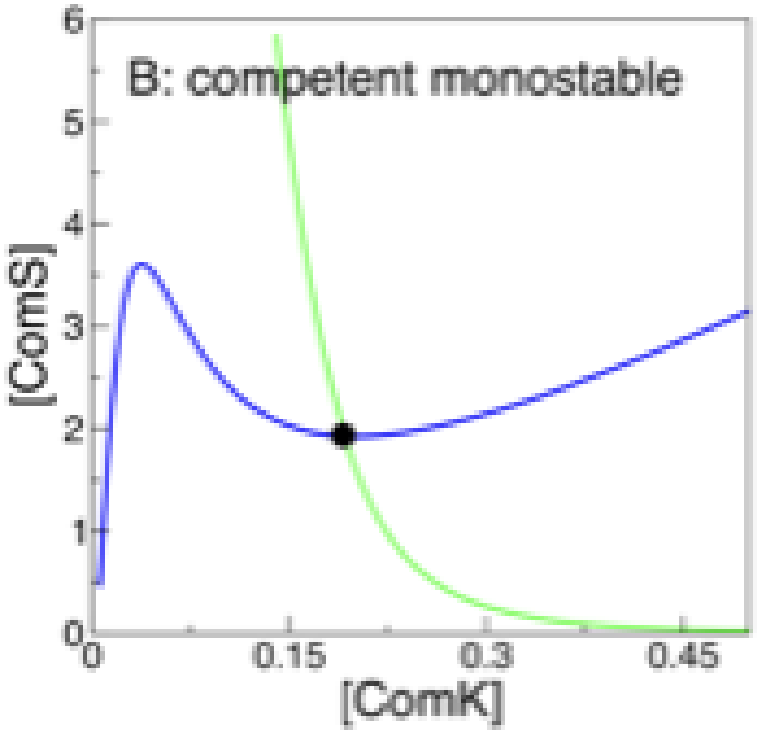
Süel GM et al, Nature (2006)

Increasing negative feedback convert bistability to the competent monostable state through another saddle-node bifurcation

$$\dot{S} = 0: \qquad S = \frac{q(K)(1+K)}{1-q(K)}, \qquad q(K) = \frac{b_s}{1+(K/k_1)^p}$$



$b_s : 0.68 \rightarrow 0.92$



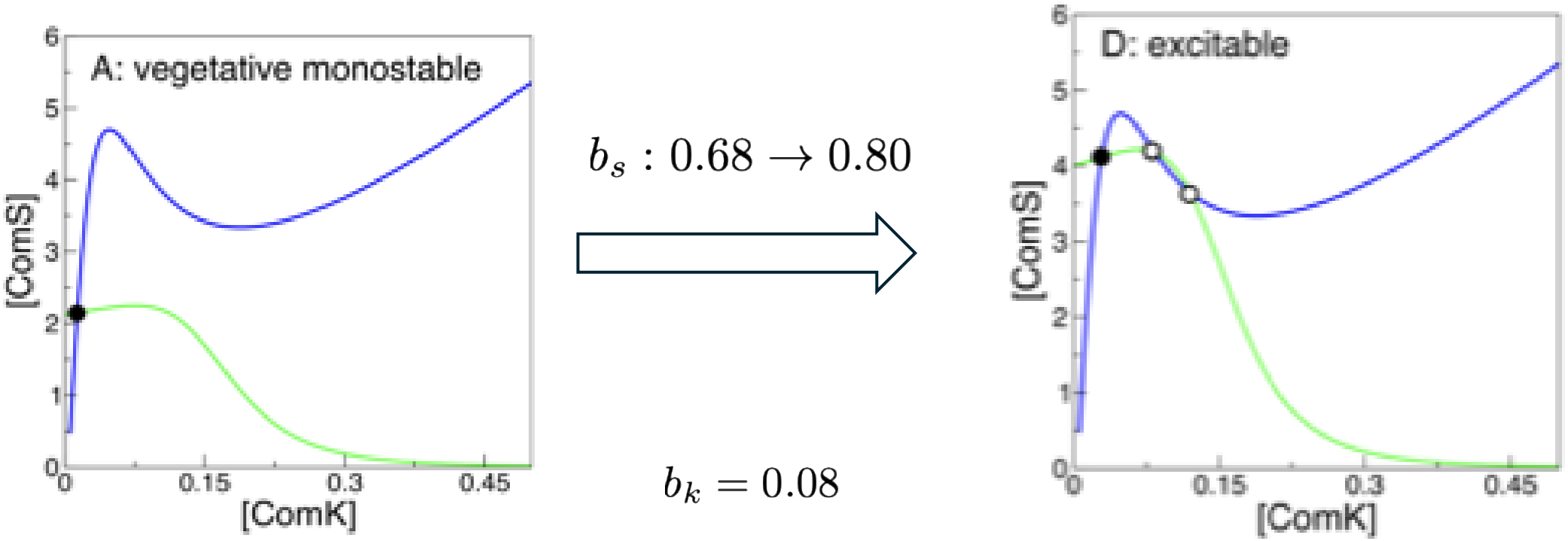
$b_k = 0.14,\ b_s = 0.68$

$b_k = 0.12, b_s = 0.92$

Süel GM et al, Nature (2006)

Some combinations of feedback strengths can generate an excitable state without creating a second stable fate

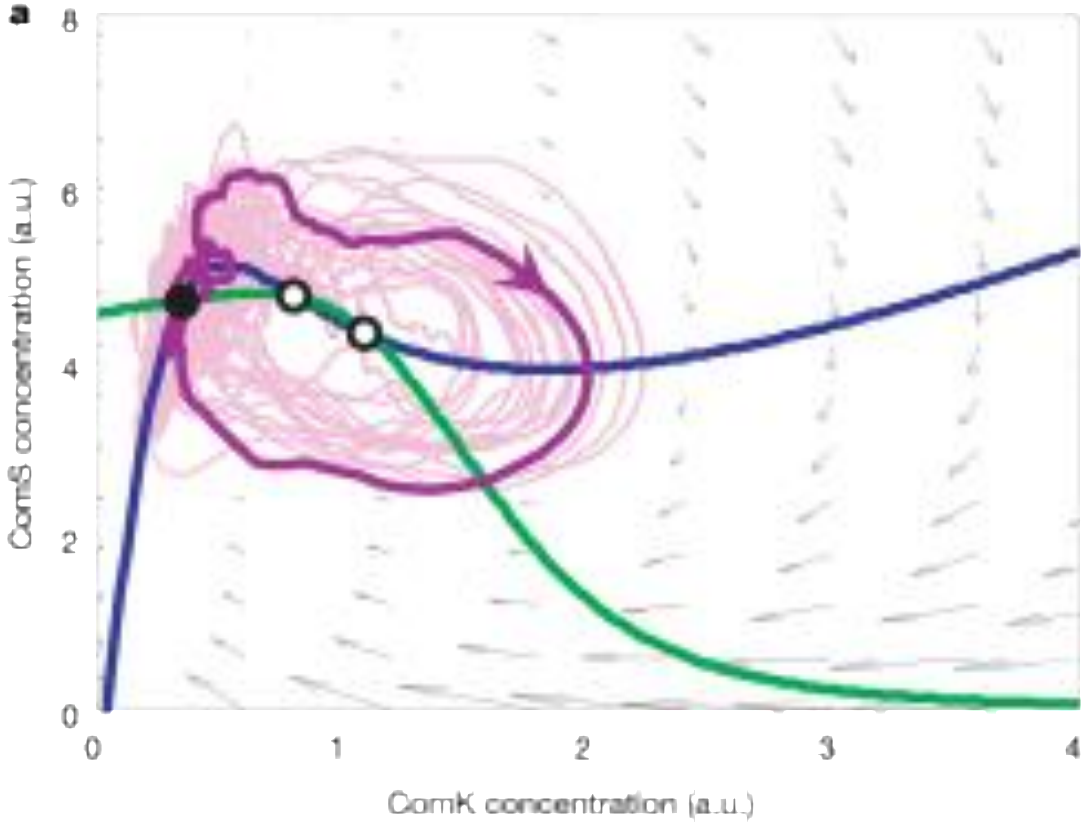
$$\dot{S} = 0: \qquad S = \frac{q(K)(1+K)}{1-q(K)}, \qquad q(K) = \frac{b_s}{1+(K/k_1)^p}$$

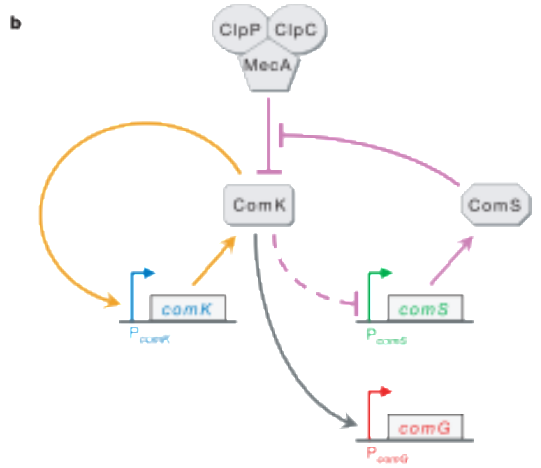


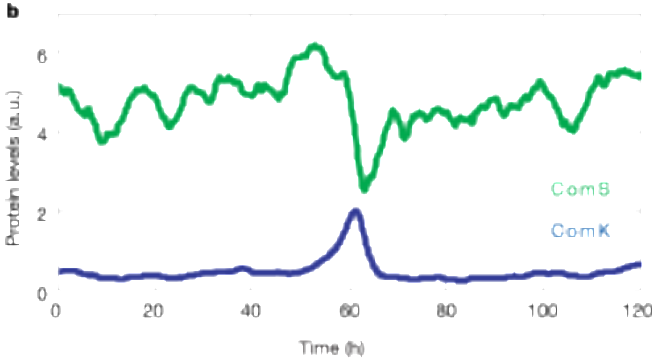
Süel GM et al, Nature (2006)

In the excitable regime, noise above the separatrix around the intermediate saddle create large excursion – transient competence

$$\frac{dS}{dt} = \frac{b_s}{1+(K/k_1)^p} - \frac{S}{1+K+S} + \xi(t)$$

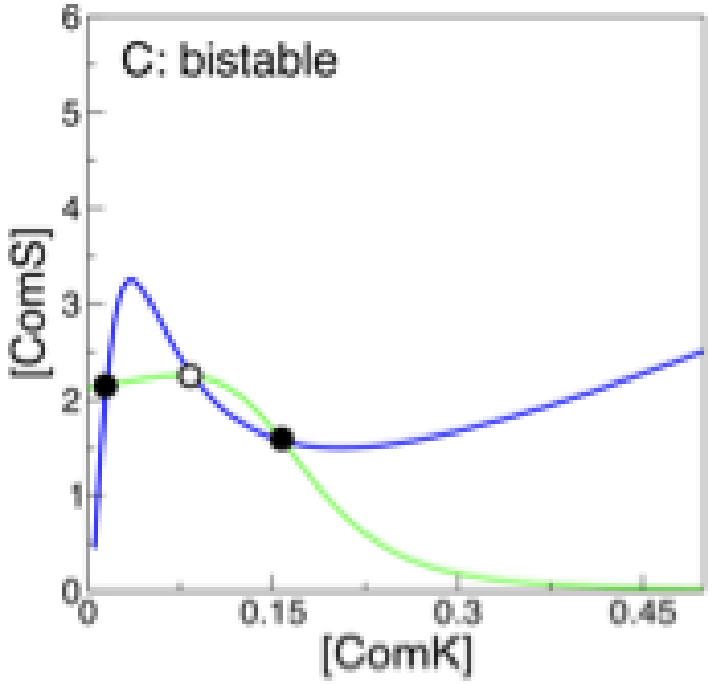


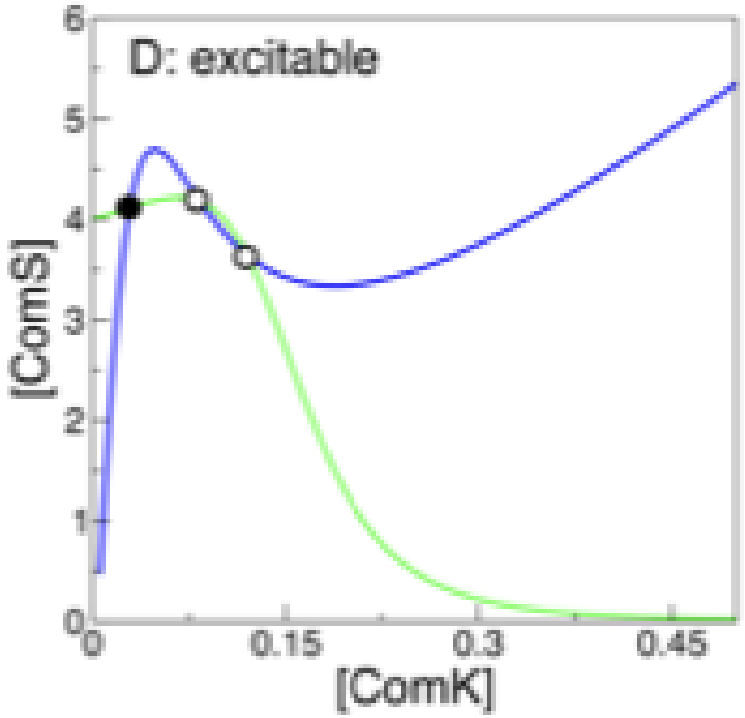




Süel GM et al, Nature (2006)

Bistability and excitability can have the same number of fixed points but different stability – Hopf bifurcation



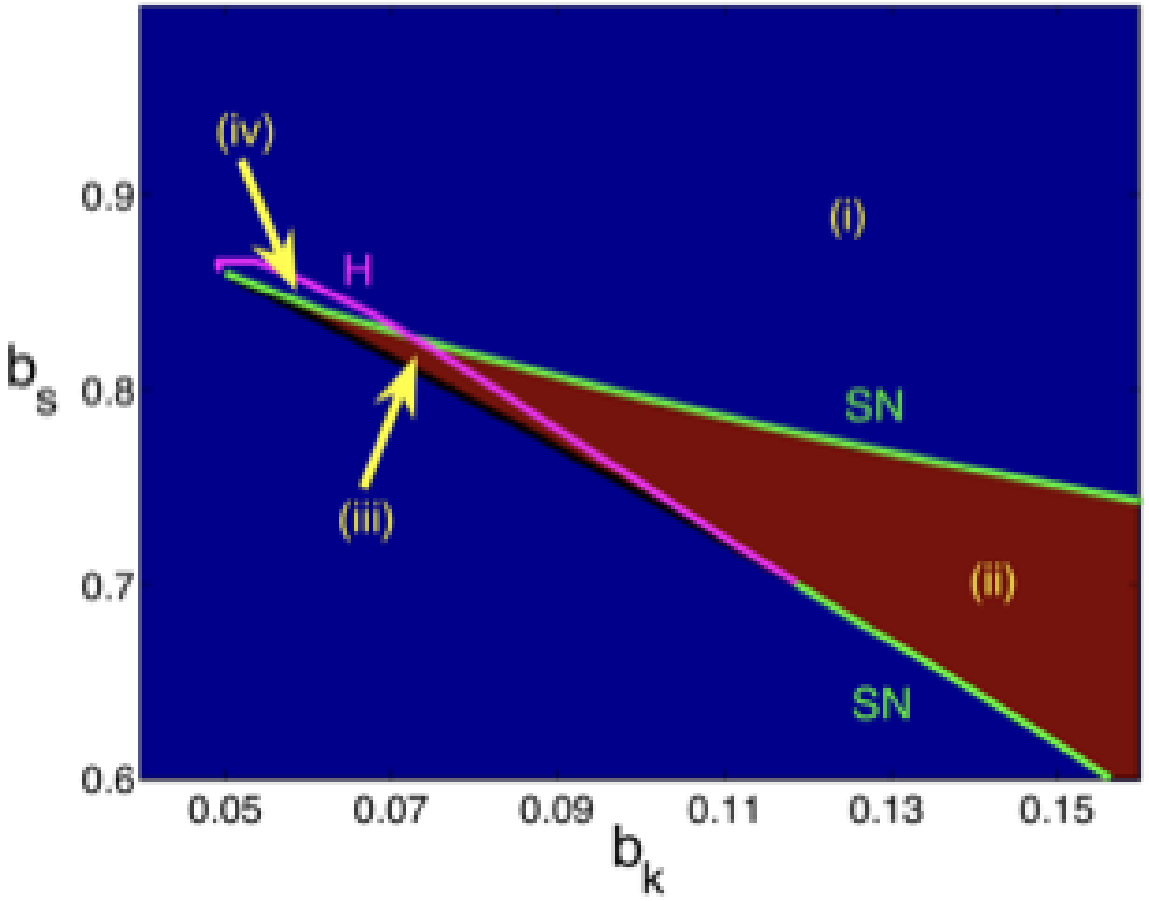


$b_k = 0.14,\ b_s = 0.68$

$b_k = 0.08,\ b_s = 0.80$

Süel GM et al, Nature (2006)

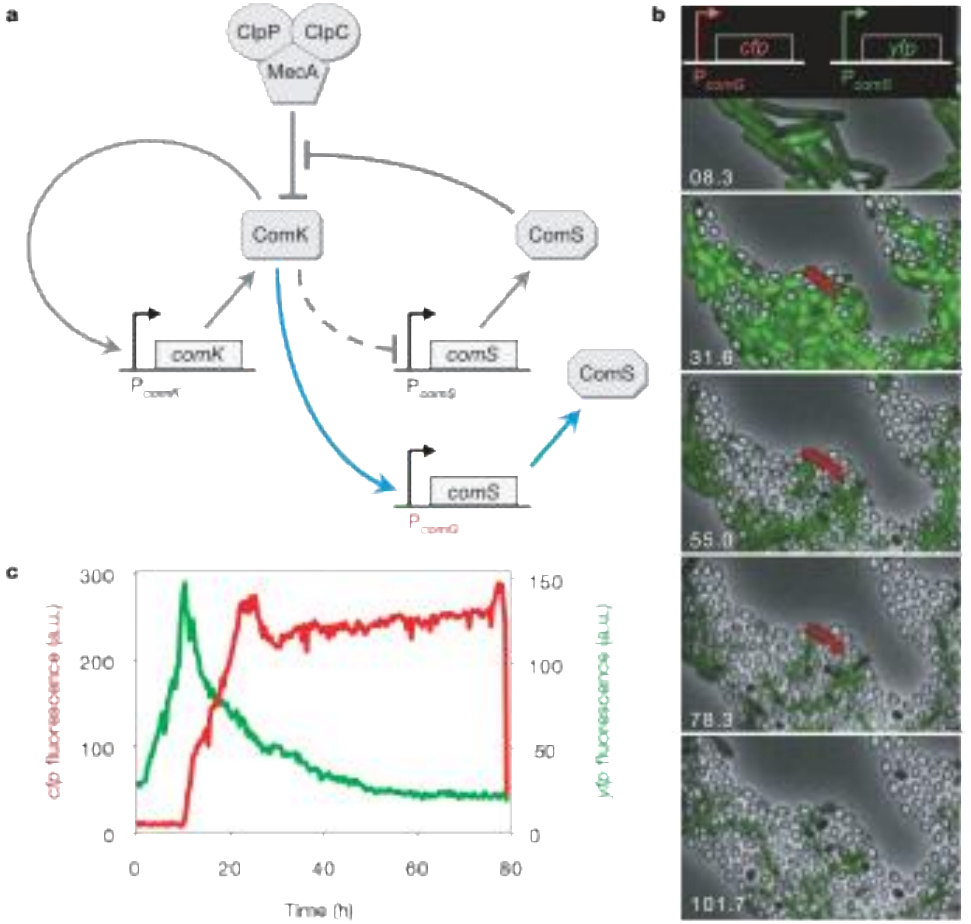
Phase portrait summarize the different dynamic regimes and bifurcations



Süel GM et al, Nature (2006)

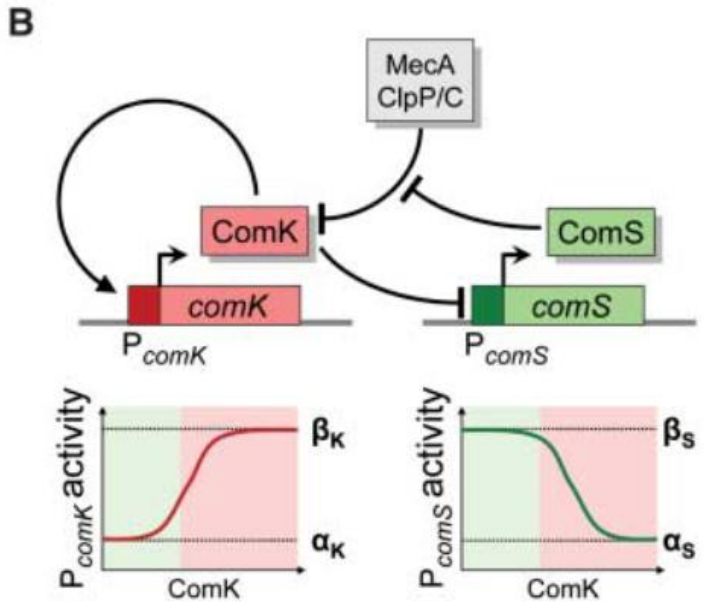
How do we know that the model is right?
Design new experiments

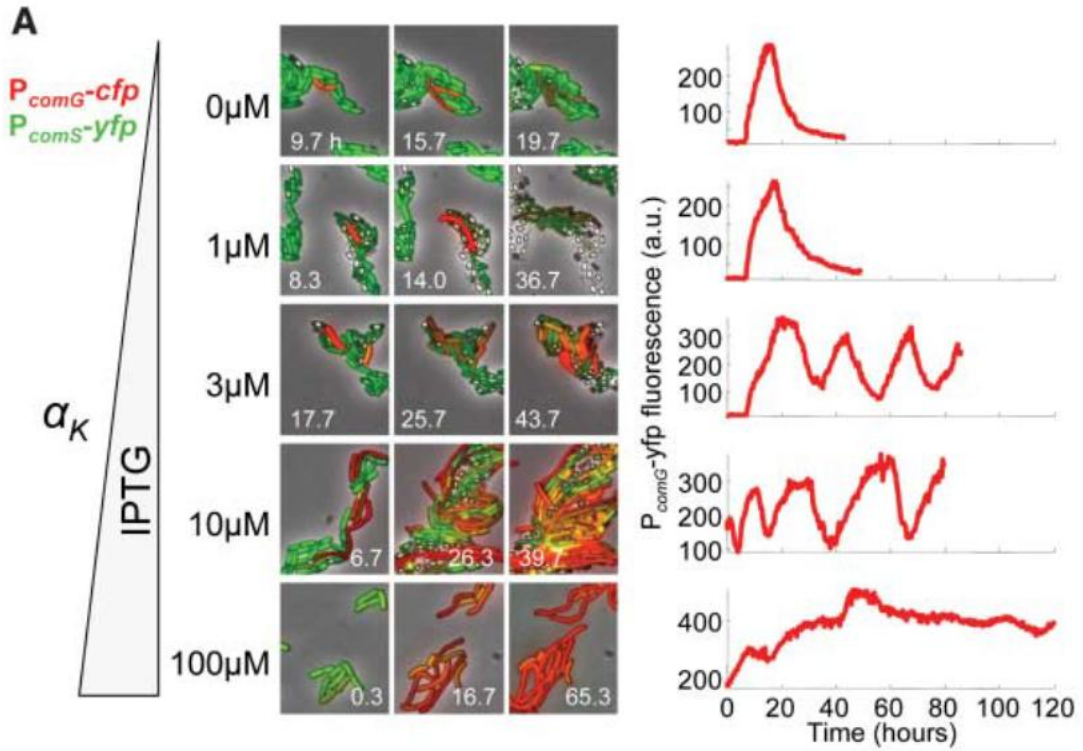
The authors constructed a synthetic compensatory circuit to demonstrate the critical role indirect negative feedback in competence exists



Süel GM et al, Nature (2006)

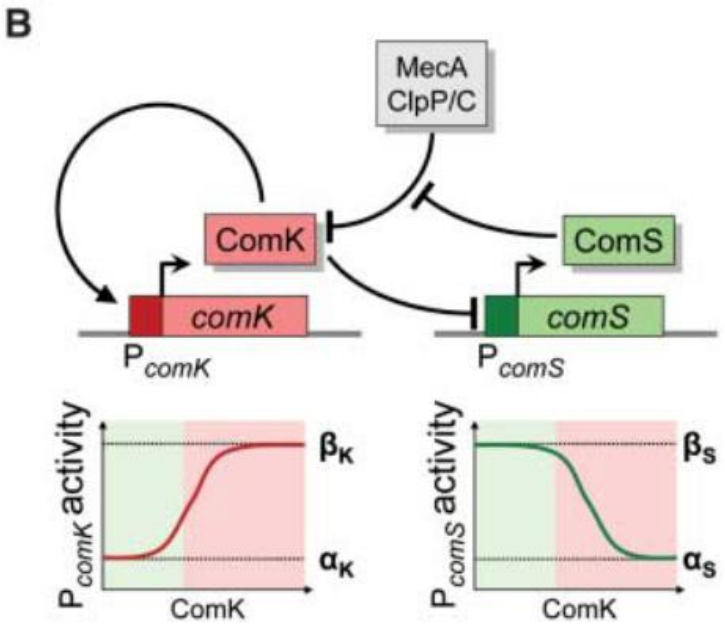
The authors showed that comK basal expression can tune initiation frequency

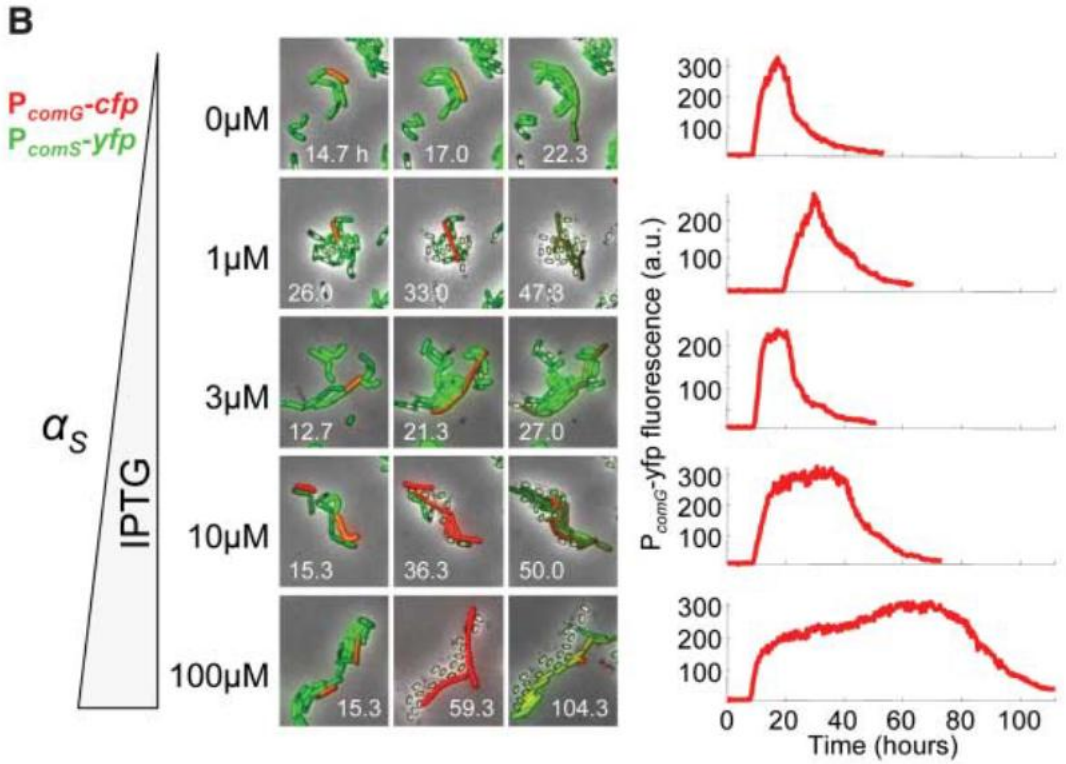




Süel GM et al, Science (2007)

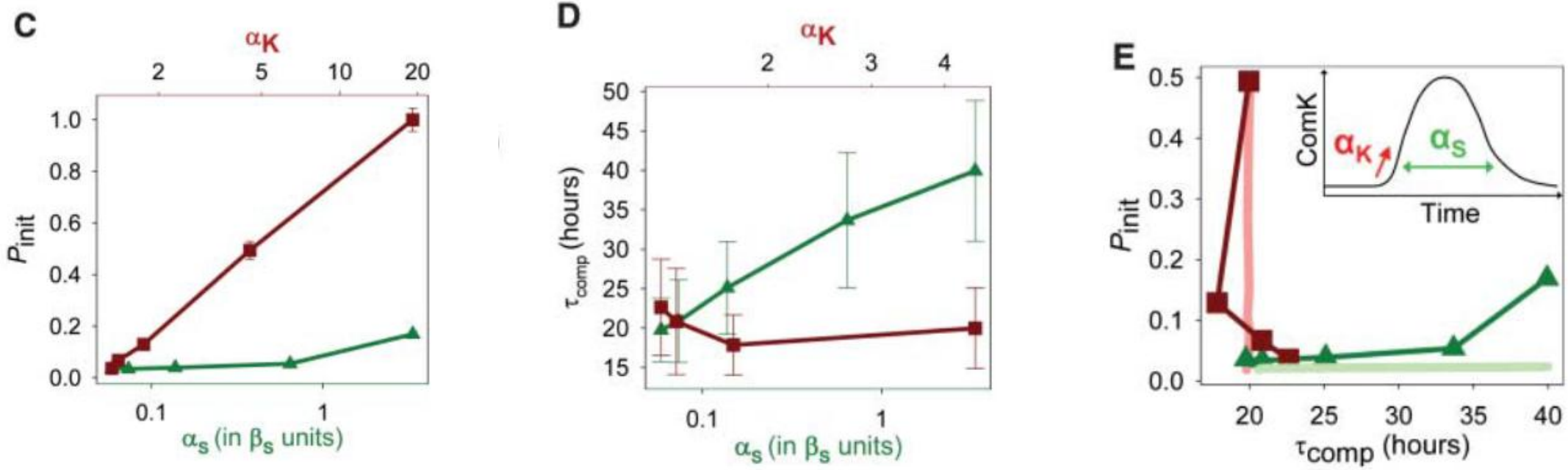
While comS basal expression can tune competence duration





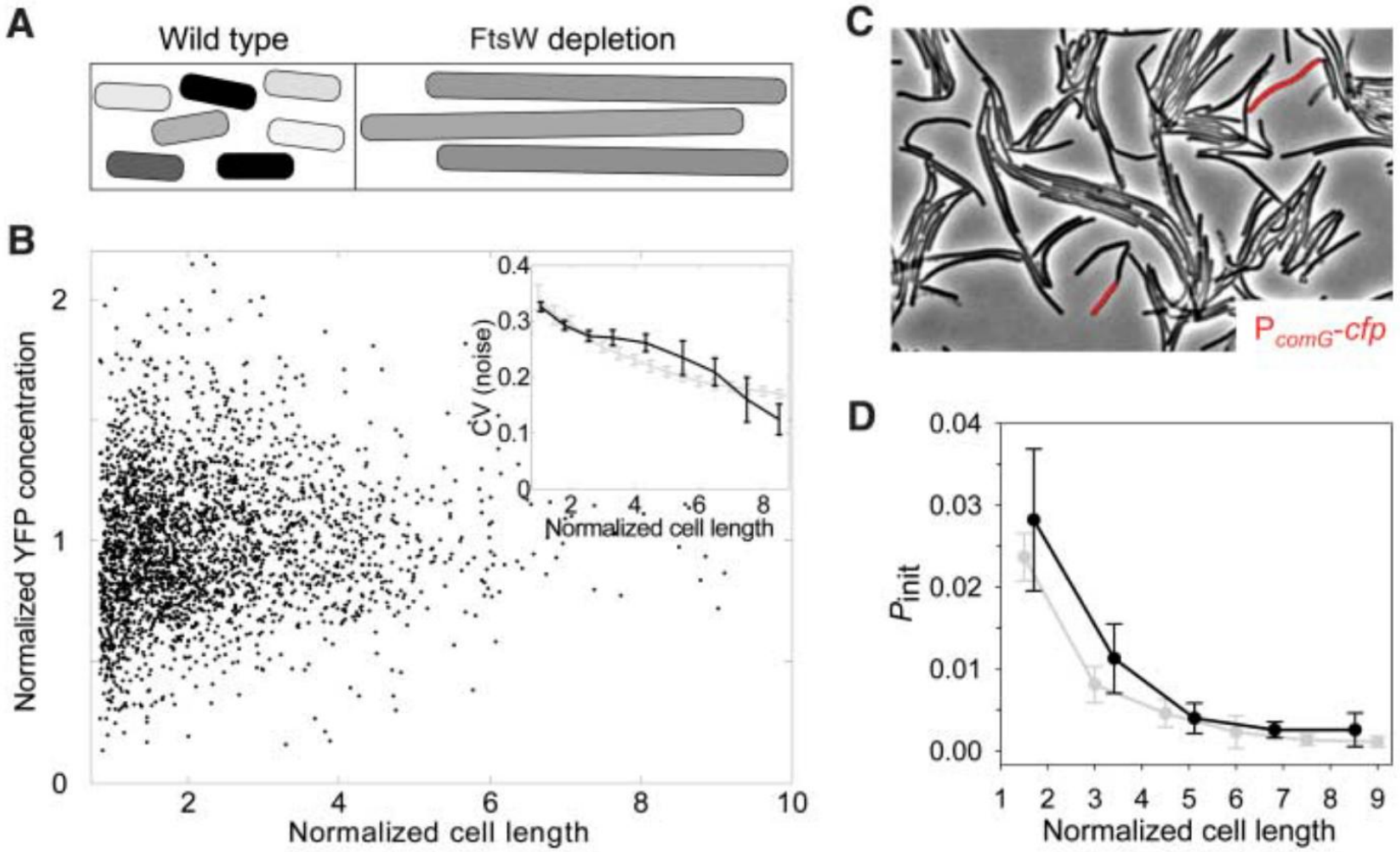
Süel GM et al, Science (2007)

And competence initiation frequency and duration can be independently tuned



Süel GM et al, Science (2007)

Reducing noise by increasing cell volume decreases initiation frequency



Süel GM et al, Science (2007)